

Go through the following search and sorting Algorithms , Answer the following questions...

Searching Algorithms

Linear Search

Binary Search

Sorting Algorithms

Bubble Sort

Selection Sort

Insertion Sort

Merge Sort

Quick Sort

Heap Sort

Counting Sort

Radix Sort

Note :

For every algorithm, you should be able to:

Explain how it works clearly.

Write the core logic or at least the main part of the code implementation.

Time and space complexity is must .

Which sorting algorithm would you choose:

- a) Nearly sorted array
- b) Large dataset
- c) Memory constraint system

Why Merge Sort time complexity is $O(n \log n)$. Is it stable? Why?

Can Binary Search work on Linked List?

Why does Quick Sort perform poorly on already sorted arrays?

Why merge sort is not in-place?

Can Quick Sort be made stable?